

04_Feature Engineering

Objective

The objective of feature engineering was to transform raw transactional data into meaningful business metrics that could support operational analysis, customer experience evaluation, KPI development, and dashboard reporting.

New analytical features were created using existing order lifecycle timestamps to provide deeper insights into fulfillment performance and marketplace operations.

Feature Engineering Strategy

The Orders table contains multiple timestamps representing different stages of the order lifecycle.

By calculating the time differences between these events, new business-oriented features were created to measure:

- Approval efficiency
- Processing efficiency
- Shipping performance
- Fulfillment speed
- Delivery quality
- Customer purchasing behavior

These features provide significantly more analytical value than the original raw timestamps.

Engineered Features

approval_days

Measures the time required to approve an order after purchase.

Formula

```
order_approved_at  
-
```

```
order_purchase_timestamp
```

Business Purpose

Evaluates order approval efficiency and internal processing responsiveness.

carrier_days

Measures the time required to transfer an approved order to the logistics carrier.

Formula

```
order_delivered_carrier_date  
-  
order_approved_at
```

Business Purpose

Evaluates warehouse and fulfillment processing efficiency.

delivery_days

Measures shipping duration after carrier pickup.

Formula

```
order_delivered_customer_date  
-  
order_delivered_carrier_date
```

Business Purpose

Evaluates logistics and shipping performance.

total_fulfillment_days

Measures the complete order lifecycle from purchase to customer delivery.

Formula

```
order_delivered_customer_date  
-  
order_purchase_timestamp
```

Business Purpose

Provides an end-to-end fulfillment performance metric.

delivery_delay_days

Measures whether an order was delivered earlier or later than expected.

Formula

```
order_delivered_customer_date  
-  
order_estimated_delivery_date
```

Business Purpose

Evaluates delivery promise accuracy and customer experience.

purchase_year

Extracted from the order purchase timestamp.

Business Purpose

Supports yearly growth and trend analysis.

purchase_month

Extracted from the order purchase timestamp.

Business Purpose

Supports monthly trend and seasonality analysis.

purchase_quarter

Extracted from the order purchase timestamp.

Business Purpose

Supports quarterly performance reporting.

purchase_weekday

Extracted from the order purchase timestamp.

Business Purpose

Supports day-of-week purchasing behavior analysis.

Feature Summary

Feature	Category
approval_days	Operational
carrier_days	Operational
delivery_days	Logistics
total_fulfillment_days	Fulfillment
delivery_delay_days	Customer Experience
purchase_year	Time Intelligence
purchase_month	Time Intelligence
purchase_quarter	Time Intelligence
purchase_weekday	Time Intelligence

Validation

All engineered features were validated using:

- Sample record inspection
- Descriptive statistics
- Logical consistency checks
- Order lifecycle verification

The generated values aligned with expected business behavior and were approved for analytical use.

Outcome

A total of **9 business-oriented features** were successfully engineered and added to the Orders dataset.

These features transformed raw operational timestamps into actionable analytical metrics and will support:

- SQL analysis
- KPI development
- Hypothesis testing
- Operational performance evaluation

- Power BI dashboarding

The enhanced dataset was subsequently prepared for PostgreSQL integration and advanced business analysis.